

PINNs on Ice

Modelling Surface Melt Dynamics in the Greenland Ice Sheet using
Physics-Informed Neural Networks

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Master's Degree Thesis in
AI for Health
Spring Term 2026
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Note for supervisor: Changes in this draft

- This draft has new sections: Abstract and Discussion.
- The plots now retain resolution at maximum zoom.
- The front matter now contains lists of figures, tables, abbreviations, and an Acknowledgement page.
- The title page now resembles the SU template.

Abstract

Introduction The Greenland Ice Sheet (GrIS) is currently undergoing accelerating mass loss, predominantly driven by surface meltwater production, representing a critical contributor to global sea-level rise. High-resolution numerical methods can resolve complex glacier dynamics, but are limited in scale. Purely data-driven models are not as limited by scale, but fail to adhere to the laws of physics, potentially resulting in poor generalization out of distribution. This research addresses this limitation by investigating the integration of thermodynamic partial differential equation (PDE) constraints within a Physics-Informed Neural Network (PINN) framework to achieve both computational efficiency and physical validity.

Background Surface melt in an ice sheet is fundamentally dictated by the Surface Energy Balance, which accounts for fluxes in radiation, and in sensible and latent heat exchange. A key boundary in the thermodynamics is at 0°C , where surplus energy ceases to raise the ice temperature and is instead consumed as latent heat of fusion for a change in phase. Traditional deep learning architectures function as unconstrained black boxes that routinely fail to recognize this physical ceiling during peak summer atmospheric forcing. While standard PINNs enforce mass and momentum conservation via automatic differentiation to solve mechanical ice flow problems, the explicit coupling of SEB thermodynamics within a composite loss landscape remains unexplored.

Methodology This study develops a hybrid Long Short-Term Memory Physics-Informed Neural Network (LSTM-PINN) trained on data from the Programme for

Monitoring of the Greenland Ice Sheet (PROMICE). The LSTM accounts for the thermal inertia and cumulative energy history. A layer with integrated physics treats bulk heat transfer coefficients as learnable parameters. A composite loss function is used to penalize the model both for deviations from sensor observations, and for straying away from the physics equations.

Results Evaluation across distinct latitudinal and elevational regimes demonstrates that the LSTM-PINN achieved an overall Root Mean Square Error (RMSE) of 1.11°C while maintaining high spatial generalization ($R^2 > 0.95$). By enforcing the thermodynamic loss penalty, the architecture successfully learned the 0°C phase-change threshold, eliminating the unphysical surface temperature overshoots characteristic of unconstrained baselines. A distinct spatial dichotomy emerged: active southern ablation zones exhibited exceptional statistical precision but elevated physical stress at the melting boundary, whereas stable northern and interior regimes demonstrated near-perfect physical adherence accompanied by marginal statistical noise induced by extreme-cold sensor drift. Ensemble analysis confirmed that the physics-based regularizer yields highly reproducible predictive envelopes regardless of random weight initialization.

Discussion The integration of differentiable SEB constraints effectively enforces thermodynamics, forcing the neural network to reconcile predictive accuracy with the physical reality of latent heat consumption. The physical violation rates observed in highly dynamic ablation zones reflect the numerical complexity of mapping a discontinuous phase change rather than systemic model failure. During instrumental telemetry blackouts, the model's capacity to transition into a purely physical emulation mode by widening its predictive confidence envelope while maintaining continuous SEB closure proves highly valuable for cryospheric data recovery. These findings validate the hypothesis that embedding thermodynamic PDE losses into deep learning models prevents non-physical extrapolation, laying the groundwork for fully coupled, fast, and physically consistent neural ice sheet models.

Acknowledgement

The authors express their heartfelt and sincerest gratitude to their supervisor, Victoria Oberländer for her constant and constructive support and guidance. The insights received were invaluable for the smooth and timely completion of this work. We thank our peers for their reviews and insights, which helped us hone and perfect this thesis. Finally, we thank our families for their endless love and support during the entirety of this project.

Abbreviations

Abbreviation	Description
PINN	Physics-Informed Neural Network
PDE	Partial differential equation
GrIS	Greenland Ice Sheet
SEB	Surface Energy Balance
RCM	Regional Climate Model
ANN	Artificial Neural Networks
PINNICLE	Physics-Informed Neural Networks for Ice and CLimateE
AWS	Automated Weather Station
PROMICE	Program for Monitoring of the Greenland Ice Sheet
LSTM	Long Short-Term Memory
RMSE	Root Mean Square Error
PVR	Physics Violation Rate
GPU	Graphical Processing Units

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1 Introduction

The Greenland Ice Sheet is losing ice faster than it gains it, and this loss is an important driver of global sea-level rise. Much of this loss is linked to melt: snow and ice melt at the surface, meltwater flows through and under the ice, and warm ocean water melts ice at glacier fronts. While global climate models can project large-scale trends in temperature and precipitation, they do not resolve the complex, multi-scale processes that control how atmospheric and oceanic forcing translate into melt, runoff, and ultimately ice-sheet thinning. Regional climate models and high-resolution ice-sheet models fill part of this gap, but their computational cost limits their use in large ensembles and strongly constrains our ability to rigorously explore model uncertainty. At the same time, the rapid growth in remote sensing and in situ observations has created an unprecedented volume of data that is not yet fully exploited in physically consistent modeling frameworks.[1]

Melting plays a crucial and complicated role in the dynamics of glaciers. Melting can lead to deformation of glaciers through pressure, fracturing of ice, and lubrication of the bed of the glacier. Melting in glaciers can occur on the top surface, at the base, and even internally. A myriad of phenomena can cause melting in glaciers, including friction between moving surfaces, absorption of solar radiation, and pressure and strain.

There is a growing abundance of heterogeneous Greenland datasets that are difficult to assimilate within traditional model frameworks. Satellite altimetry provides detailed records of surface elevation change, while optical and radar imagery offer maps of surface velocity and calving front positions. Ice-penetrating radar and compilations such as BedMachine give estimates of bed topography and ice thickness, albeit with spatial gaps and uncertainties, especially near fast-flowing outlets. Regional climate models provide gridded estimates of surface mass balance and surface melt, and ocean reanalyses or regional ocean models supply thermal forcing at marine glacier fronts. Integrating these disparate data sources into a consistent estimate of melt and flow would usually require complex data assimilation schemes and repeated forward runs of computationally expensive models.

In this context, machine learning has emerged as a powerful tool to emulate components of the ice-sheet system and to extract patterns from large datasets. For Greenland, neural-network-based emulators have already been used to approximate surface melt calculated by detailed surface mass balance schemes, as well as to mimic the behavior of sophisticated subglacial hydrology models. These

emulators can reproduce complex nonlinear relationships between forcing and response at a fraction of the computational cost of the original models, enabling rapid scenario exploration and probabilistic assessments. However, most existing applications treat the learning problem as purely data-driven: the networks are trained to fit outputs from a physical model or observations, but they are not explicitly constrained to satisfy the underlying conservation laws or governing equations. As a result, such models may generalize poorly outside the training distribution, and they provide limited physical interpretability, especially when extrapolating to future climate states.

Physics-informed neural networks (PINNs) offer a promising way to bridge this gap between data-driven learning and physically consistent modeling.[2] Instead of relying solely on observational or model-generated labels, PINNs embed the governing partial differential equations (PDEs) directly into the training objective.[3] The neural network is trained not only to match available data, but also to minimize the residuals of the PDEs, boundary conditions, and initial conditions. This approach allows PINNs to act as mesh-free solvers or emulators that can incorporate observational constraints, while still respecting mass, momentum, and energy conservation. In glaciology, recent work has demonstrated the potential of PINNs to solve forward and inverse ice-flow problems, to infer basal friction or bed topography from surface observables, and to enforce mass conservation when interpolating ice thickness. These early studies show that PINNs can recover key dynamic fields with competitive accuracy compared to traditional numerical methods, and that they can seamlessly integrate diverse datasets into a unified inversion framework.

Despite this progress, the explicit modeling of melt dynamics within a physics-informed neural framework remains relatively unexplored, especially for the Greenland Ice Sheet. Most PINN applications in cryospheric science focus on kinematic aspects of the ice, such as velocity, stress, and thickness, while treating melt as a phenomenon subsumed by the equations. Conversely, machine-learning studies that target melt — whether surface melt, basal melt, or subglacial hydrology — typically forgo PDE-level constraints in favor of purely data-driven simulation. This separation leaves an important gap: we lack models that can both represent the complex physics of melt generation and routing, and incorporate observational constraints on ice geometry and flow within a single, coherent framework. Given that meltwater production and routing strongly influence ice dynamics through changes in basal lubrication, effective pressure, and frontal ablation, this gap limits our ability to attribute observed changes in Greenland’s outlet glaciers to spe-

cific melt processes and to project future behavior under warming.

This study addresses this gap by investigating the integration of thermodynamic constraints within a physics-informed framework. The primary research question is: Do Physics-Informed Neural Networks benefit from the inclusion of a melt dynamic PDE loss term, and does this integration lead to more accurate and physically robust predictions of glacier evolution?

Given the high dimensionality of fully coupled thermo-mechanical ice flow and the extreme computational cost associated with optimizing complex PINN architectures, this study is designed as an exploratory investigation. We will focus on a heavily constrained set of thermodynamic variables and utilize a limited, idealized dataset. The objective is not to provide a comprehensive, predictive model of the Greenland Ice Sheet, but rather to evaluate the theoretical feasibility and localized predictive improvements of enforcing melt dynamic constraints within a neural network’s loss.

2 Background

2.1 The Greenland Ice Sheet and Global Climate Change

The Greenland Ice Sheet (GrIS) is — after the Antarctic Ice Sheet — the second largest mass of freshwater ice on Earth. As such, it is one of the most critical components of the Earth system, with profound implications for global sea-level rise, regional climate, and ocean circulation. Over the past few decades, observations have revealed an accelerating mass loss from Greenland driven by a combination of increased surface melt and dynamic discharge of ice through fast-flowing outlet glaciers. Computer models can simulate many of these processes, but detailed ice-sheet and climate models are very expensive to run and can be difficult to calibrate with the large amount of data now available from satellites and other observations. At the same time, simpler machine-learning models are fast but often behave like black boxes, making it hard to know whether their predictions remain physically realistic in new situations, such as future climate scenarios.[4]

2.2 Thermodynamics of Surface Melt

At atmospheric pressure, melting of ice is triggered at a temperature of $0^{\circ}C$. Further incoming energy is then absorbed by the ice for the change of phase, rather than increase of temperature. This absorbed energy is called the latent heat of

fusion for ice, L_f , and can be defined as the amount of energy required to convert ice to water at the same temperature. The value of L_f is determined to be around $3.34 \times 10^5 J kg^{-1}$. On a glacier, the amount of energy available for melting is obtained from multiple sources. Incident solar radiation (shortwave), if absorbed, warms up the ice. This amount absorbed depends on the reflective properties of ice. If α denotes the surface albedo of the ice, and SW_{in} represents the incoming shortwave, the amount of radiation absorbed by the ice would be $SW_{in}(1 - \alpha)$.

There also exists a flux of longwave radiation through the surface of the ice. Thermal energy from the atmosphere and greenhouse gases emitted downwards falls on to the ice surface. The ice surface also loses heat by emitting thermal radiation in the infrared spectrum. This flux of longwave, denoted as ΔLW , contributes significantly to the melt.

Two other forms of heat transfers also occur at the ice surface, namely the sensible heat transfer Q_S , and the latent heat transfer Q_L . Q_S is the heat transferred directly from one material to another, for example, from the glacier to the atmosphere or the other way around. This is separate from the heat transferred by emission of radiation. This heat is dependent on the difference of temperatures between the two materials, and how much they are in contact. Q_L is the energy consumed or released during phase changes of water at the ice surface. A humid atmosphere above the ice would result in the condensation of moisture on to the ice. This process releases energy – the latent heat of vaporisation – to the ice, contributing to melt. On the other hand, a dry atmosphere results in sublimation or vaporisation of water, which consumes energy from, and subsequently cools down, the glacier. [5]

With all these factors contributing to melt, the total energy available for melting can be represented as the sum

$$Q_M = SW_{in}(1 - \alpha) + \Delta LW + Q_S + Q_L \quad (1)$$

This is called the surface energy balance (SEB) equation. If at the melting point $T = 0^\circ C$, there is a positive value of the available energy, $Q_M > 0$, the energy is used up for the change of phase. The melt rate, or amount of meltwater produced, is obtained mathematically by

$$M = \frac{Q_M}{\rho L_f} \quad (2)$$

where $\rho = 1000 kg m^{-3}$ is the density of water.

2.3 Traditional Numerical Modeling and Regional Climate Models

To project glacier evolution, researchers rely on physics-based numerical modeling and purely data-driven machine learning. In traditional frameworks, surface melt is usually resolved using Regional Climate Models (RCMs).[6] However, solving the full SEB over complex glacier topography within an RCM is exceptionally computationally expensive. Consequently, RCMs are often run at relatively coarse spatial resolutions (e.g., 10 km to 20 km grids) or run offline, generating static surface mass balance fields that are subsequently fed as one-way boundary conditions into mechanical ice sheet models. This approach severely limits the ability to perform coupled, high-resolution transient simulations or rigorous uncertainty quantification.

2.4 Data-Driven Modeling and Machine Learning Emulators

To bypass the computational bottlenecks of numerical solvers, the glaciological community has increasingly adopted deep learning. Standard Artificial Neural Networks (ANNs) excel at learning complex, non-linear mappings directly from data. In glaciology, purely data-driven models have been deployed successfully for static problems, such as mass-conserving ice thickness estimation or mapping basal topography from remote sensing data.

Koo et al.[7] emulated the kinetics of the breaking off (calving) of glaciers into the sea, using graph neural networks (GNNs). They succeeded in mapping variables like ice thickness, basal friction, and velocity to predict future states of the Helheim glacier. The resulting model was 30-34 times faster than the traditional emulator at the time.

Verjans and Robel[8] developed a convolutional neural network architecture that emulates subglacial hydrology – how surface water travels downwards to the base of the glaciers. The model was trained on multiple major glaciers in Greenland, and achieved high spatial and temporal generalization scores ($R^2 > 0.95$ and $R^2 > 0.99$, respectively), while accurately capturing the seasonal transitions of the drainage system. The model reduced computational time by a factor of $\mathcal{O}(10^3)$.

2.5 Limitations of Unconstrained Deep Learning

While these models offer massive speedups, these deep learning models function as “black boxes”. Additionally, a standard neural network predicting ice velocity

or geometry does not inherently respect the conservation of mass or momentum. This lack of physical constraint makes them unreliable when extrapolating to extreme, out-of-sample future climate scenarios.

Typically, the loss function takes a form such as:

$$\mathcal{L} = \frac{1}{N} \sum_{i=1}^N \|\hat{\mathbf{y}}(x_i) - \mathbf{y}(x_i)\|^2,$$

where $\mathbf{y}(x_i)$ are the target outputs from observations, and $\hat{\mathbf{y}}(x_i)$ are predicted outputs of the model, given inputs x_i . This approach does not have explicit penalties for violating the underlying partial differential equations. When extrapolating beyond the training range, the model may produce outputs that are not in agreement with the laws of conservation of mass, energy, and momentum.

Physics-informed neural networks can be used to extend this idea by adding physics-based penalties to the loss function. The loss function can now take the form

$$\mathcal{L} = \lambda_{data}\mathcal{L}_{data} + \lambda_{PDE}\mathcal{L}_{PDE} + \lambda_B\mathcal{L}_B$$

where the λ are weighting factors. The first term addresses data misfits. The second and third terms are the weighted losses for the PDEs and the boundary conditions, respectively. This way, the model will be penalized for straying from both observational data and theoretical limits set by the PDEs.

2.6 Physics-Informed Neural Networks in Glaciology

Cheng et al.[9] applied such a framework to Helheim Glacier, using a momentum balance and thickness evolution equation as PDE constraints, and surface velocity and thickness observations as data constraints. They showed that the PINN can recover hidden fields, such as ice thickness or basal friction, from partial observations, and that it can handle both forward and inverse problems in a unified way. The work, named PINNICLE(Physics-Informed Neural Networks for Ice and CLimatE), generalizes this approach into a reusable library, providing implementations of the Shelfy-Stream Approximation (SSA) and related equations, along with tools for defining PDE residuals, sampling collocation points, and training PINNs for various ice-sheet configurations.

2.7 The Research Gap: Coupling Melt Dynamics in PINN Frameworks

To date, however, most PINN applications for ice-sheet problems focus on mechanical variables (velocity, thickness, basal drag, bed topography), and melt processes are either not represented explicitly or are treated as consequences of kinematic phenomena[10]. Conversely, neural emulators that target melt processes for Greenland do not incorporate PDE residuals in the loss and thus do not qualify as PINNs[6, 8].

Our novelty lies in bridging this gap by embedding a simplified surface melt formulation into the PINNICLE architecture. Rather than relying on static, pre-computed surface boundary conditions, we propose integrating a surface melt PDE constraint into the composite loss function alongside the mechanical flow constraints. By evaluating the theoretical feasibility of this thermo-mechanical loss landscape on a constrained dataset, we hypothesize that the network can learn to simultaneously respect the mechanics of ice flow and the thermodynamics of surface melt, paving the way for fully coupled, computationally efficient neural ice sheet models.

3 Methodology

This study develops a PINN to model and predict the surface temperature and occurrence of melt at the surface of the GrIS. The PINN’s function approximator is constrained by the SEB and the heat equation, in the interest of maintaining all predictions within the realm of thermodynamic validity. The study focuses on GrIS, utilizing data from three distinct automated weather stations (AWS) operated by Program for Monitoring of the Greenland Ice Sheet (PROMICE).

3.1 Problem Formulation

The formulation of this study relies on two fundamental decisions made — the utilization of station-based data over gridded reanalysis products, and the isolation of surface melt dynamics over a fully coupled model covering basal melt and hydrology. These constraints were deliberately chosen to ensure the physical validity and mathematical tractability of the neural network optimization process. To provide atmospheric forcing and boundary conditions, this study utilizes in situ data from automatic weather stations (PROMICE and GC-Net)[11] rather

than widely used gridded reanalysis products, such as ECMWF’s ERA5[12]. While reanalysis datasets offer the distinct advantage of spatiotemporally continuous, gap-free coverage across the entire Greenland Ice Sheet, they are fundamentally model outputs rather than direct observations. Training a Physics-Informed Neural Network on ERA5 data presents a critical epistemological problem: the network would not be learning the true physical relationship between the atmosphere and the ice. Instead, it would merely be learning to emulate the underlying parameterizations and inherent biases of the ERA5 assimilation model. By relying exclusively on high-frequency, station-based PROMICE data, the PINN is forced to reconcile its internal thermodynamic equations against actual, localized physical reality. This ground truth approach guarantees that the resulting network weights reflect real-world energy partitioning, which is essential for proving the efficacy of the physical loss constraint.

Glacier melt is a multi-faceted process encompassing surface ablation, internal strain heating, subglacial frictional melting, and marine-terminating frontal melt. However, this study restricts its thermodynamic scope entirely to surface melt driven by the SEB. The decision to exclude basal melt and subglacial hydrology was driven by both glaciological significance and computational feasibility.

From a mass balance perspective, surface melt and subsequent runoff have become the dominant drivers of mass loss for the Greenland Ice Sheet in the 21st century, regularly outpacing dynamic ice discharge. Accurately resolving the dynamics of the SEB is therefore the most pressing priority for predictive glaciology.

Second, from a machine learning perspective, integrating subglacial hydrology and basal melt into a PINN introduces a certain degree of uncertainty. Basal hydrology on its own is a complicated system involving several variables that cannot be directly measured, such as geothermal heat flux and exact basal water pressure. Attempting to train a neural network to study basal hydrology along with surface ablation with any significant utility would require an amount of data large enough to be computationally too expensive.

By isolating the surface melt boundary condition, this study provides a constrained, heavily observed, and highly impactful testbed. It allows for the evaluation of integrating a thermodynamic PDE into a neural network’s loss function without the confounding noise of unobservable basal parameters, serving as a necessary precursor to future, fully coupled thermo-mechanical PINN architectures.

3.2 Data Acquisition and Station Selection

To ensure the PINN model captures a generalized representation of the GrIS thermodynamics, ten automated weather stations were selected from the PROMICE network. The selection is tabulated in Table 1 and represented on the map of Greenland in Figure 1. These stations were strategically chosen to provide a comprehensive cross-section of the ice sheet’s diverse climatic and physical regimes, spanning from 2016 to present time. The justification for this specific configuration is rooted in three primary scientific objectives:

Latitudinal Gradient Exploration: The selection spans the entire latitudinal range of the ice sheet, from the southern margins (e.g., QAS_L at approximately 61°N) to the high-Arctic northern regions (e.g., KPC_L at 79°N). This distribution forces the model to learn thermodynamic consistency across vastly different solar radiation and temperature regimes.

Elevational Lapse Rate Modeling: By utilizing paired stations at different altitudes within the same region (e.g., KAN_L at the lower margin and KAN_U at the higher inland position), the architecture is trained to resolve the vertical lapse rate. This ensures the model accurately predicts how T_{surf} and melt dynamics change as a function of elevation.

Interior Accumulation Baseline: The inclusion of EastGRIP (EGP) serves as a critical control for the stable interior. Positioned in the accumulation zone, EGP provides a baseline for the model to distinguish between the highly active, melting ablation zones of the coast and the stable, cold interior of the ice sheet.

By integrating data from these ten locations, the dataset represents a robust spatial-temporal sample that allows the LSTM-PINN architecture to transcend localized correlations and move toward a universal physical representation of ice-sheet melt dynamics.

3.2.1 Justification for Limited Northern Station Density

While the station network includes high-latitude sites such as THU_L and KPC_L to act as northern thermodynamic anchors, the total density of stations in the extreme North of Greenland is intentionally lower than in the southern and western sectors. This apparent bias in selection is justified by two primary factors.

Figure 1: Geographical representation of Weather Station locations



Table 1: AWS from where data was sourced

#	Station ID	Longitude (°W)	Latitude (°N)	Location
1	KPC_L	-19.601	79.911	Northeast Greenland
2	THU_L	-68.266	76.399	Northwest Greenland
3	EGP	-35.500	75.600	Central interior
4	UPE_U	-52.703	72.887	West Greenland
5	SCO_L	-27.233	72.223	East Greenland margin
6	KAN_L	-50.067	67.095	West Greenland margin
7	KAN_U	-47.017	67.000	West Greenland interior
8	TAS_L	-38.899	65.640	Southeast Greenland margin
9	QAS_U	-46.233	61.175	South Greenland
10	QAS_L	-46.849	61.030	South Greenland

First was continuity in the data and the reliability of the instruments. Stations in the High Arctic (latitudes $> 75^\circ\text{N}$) are subject to extreme environmental conditions, including prolonged polar nights and severe wind-loading. These factors contribute to a higher frequency of sensor failure and battery depletion within the study period. To maintain the 10-year temporal integrity required for the LSTM-PINN’s training, several northern stations were omitted due to significant data gaps that would have necessitated excessive interpolation, thereby compromising the physical accuracy of the model.

Second was the reduction of signal-to-noise ratio in the melt data. The primary objective of this research is the modeling of melt dynamics. In the far North, surface temperatures rarely exceed the 0°C threshold for extended periods, resulting in a negligible melt signal. Including a high density of these “no melt” sites would potentially bias the neural network toward a trivial solution, thereby reducing its sensitivity to the complex energy balance transitions and phase-change dynamics occurring in the high-melt ablation zones of the South and West.

3.3 Data Preprocessing

Initial exploratory data analysis revealed severe, multi-year horizontal data gaps prior to 2016, caused by hardware failures or harsh winter conditions. While short gaps can be mathematically interpolated, interpolating multi-year weather blackouts would force the network to learn incorrect thermodynamic laws. The temporal domain encompasses a continuous period from 2016 to the present day.

It is in this range that all sensors from the chosen stations had continuous and simultaneous operational telemetry.

Raw PROMICE data was ingested into the model’s computation from the GEUS THREDDS Data Server (<https://thredds.geus.dk>). To prevent noise from high-dimensional data and to avoid overfitting, the input feature vector was pruned to include only variables that were reasoned to be relevant to the project. Variables relating to station metadata, battery voltages, and redundant downward radiation sensors were discarded. The remaining variables represent irreducible thermodynamic drivers required to calculate the surface boundary conditions. These are

Spatiotemporal variables time, gps_lat, gps_lon, and gps_alt were used respectively as coordinates (t, x, y, z).

Thermodynamic variables Air temperature (t_u), relative humidity (rh_u), wind speed (wspd_u), and albedo (albedo).

Before being used to train the neural network, the data is passed through a pre-processing pipeline to handle missing values and the inherent scaling disparities between the variables. The network relies on a combination of spatiotemporal coordinates and atmospheric variables to predict the surface temperature of the ice sheet. The spatial coordinates include GPS latitude, longitude, and elevation. The elevation data (gps_alt) underwent specific filtering to remove abrupt spikes caused by temporary GPS signal loss or station repositioning. Analysis of the raw spatial telemetry revealed severe corruption in the GPS altitude (gps_alt) data. Stations KAN_L and KAN_M suffered from hardware defaults where missing location data was erroneously recorded as exactly 0.0 meters. Conversely, the high-altitude EGP station exhibited extreme vertical triangulation spikes of up to 100 meters day-over-day due, perhaps owing to poor satellite geometry. For lower altitude stations (KAN_L and KAN_M), a threshold (readings below 100m, for example) was set to mark unphysical altitude changes. Altitude changes that crossed this threshold were converted to NaN. For the summit station (EGP), a rolling median filter was applied to eliminate satellite triangulation spikes while preserving the true macro-trend. The resulting NaN gaps were filled with linear interpolation. Linear interpolation for such short gaps in spatial data is justified owing to the the slow speed of spatial coordinate changes in ice sheets and glaciers. In nature, these changes are slow and continuous, thus, drawing a linear mathematical line between known coordinates perfectly mimics physical reality.

The cyclical pattern of the calendar was not apparent in the sequential representation of dates in the data. Time, therefore, had to be transformed to capture

the seasonal periodicities of weather. This was achieved using sine and cosine transformations of the day of the year, resulting in two variables:

$$\text{day_sin} = \sin\left(\frac{2\pi \cdot \text{day_of_year}}{365.25}\right) \quad (3)$$

$$\text{day_cos} = \cos\left(\frac{2\pi \cdot \text{day_of_year}}{365.25}\right), \quad (4)$$

where `day_of_year` stood for the number of the day in a 365-day cycle.

The atmospheric drivers (`t_u`, `rh_u`, `wspd_u`) suffered from intermittent, short-term outages due to sensor freezing or extreme storms. These gaps were imputed using spline interpolation. Unlike linear interpolation, splines fit a mathematical curve to the data. Because daily weather fluctuates in non-linear waves (e.g., diurnal temperature cycles), polynomial splines provide a highly accurate representation of the missing physical states without artificially flattening the energy peaks.

Albedo proved to be the most frequently missing variable, suffering from over 2,000 missing days per station. During polar nights, owing to lack of incident solar radiation, the dataset would not have a reading for this variable. The value of this variable is set to 0.85 for these periods. In glaciology, 0.85 is the scientifically established baseline for fresh, dry polar snow. In the SEB (equation 1), the lack of sunlight is reflected in the low value of the shortwave radiation term.

3.4 Data Architecture

The defining architectural feature of this methodology is the treatment of the target variable, `T_surf`. In the raw dataset, `T_surf` was missing for a combined total of 10,837 days across all stations. A standard supervised learning pipeline requires the removal of all rows lacking a target label. However, in this PINN architecture, these 10837 rows were intentionally retained. A new boolean flag column, `T_surf_is_collocation`, was generated. When this flag is `False`, the network calculates its statistical error against the known sensor data. When this flag is `True`, the network operates in an unsupervised mode. These coordinates become Collocation Points — points in space and time where the model is forced to predict `T_surf` relying entirely on the surrounding weather data and the hard-coded laws of physics (the SEB equation) to deduce the missing temperature.

To recapitulate, the Surface Energy Balance equation is mathematically written as

$$Q_M = SW_{in}(1 - \alpha) + \Delta LW + Q_S + Q_L,$$

where

- $SW_{in}(1 - \alpha)$ is the term that describes the shortwave flux through the ice surface.
- The second term, ΔLW , describes the flux of longwave radiation through the surface.
- The terms Q_S and Q_L stand for the sensible and latent heat fluxes through the surface.

The amount of energy absorbed, in the shortwave flux term, is influenced by albedo (α), which is obtained from the albedo variable in the data. The latent heat fluxes are dependent on, and therefore drive by, the atmospheric temperature, humidity, and wind speed. By providing the network with T_u , rh_u , $wspd_u$, and albedo, the PINN is equipped with the exact parametric variables required to approximate these physical fluxes intrinsically. The data engineering pipeline yields a finalized dataset ready for analysis, devoid of physical anomalies and missing values (NaN). The schema of the final dataset is structured as illustrated in Table 2.

3.5 Model Architecture: LSTM-PINN

The core architecture employed in this study is a hybrid PINN that utilizes Long Short-Term Memory (LSTM) cells as its primary temporal backbone. This design was necessary to accommodate for the fact that surface melt is not merely a function of the instantaneous weather conditions, but also dependent on cumulative energy fluxes over time.

3.5.1 Temporal Feature Extraction (Stacked LSTM)

The initial component of the architecture consists of a two-layer stacked LSTM. The model processes input features in 7-day sliding windows, allowing the network to capture seasonal transitions and multi-day heatwave signatures. A dropout rate of 20% is applied between the LSTM layers to mitigate overfitting to sensor noise and improve spatial generalization across the ten stations.

3.5.2 Prediction Head and Activation

The final hidden state of the LSTM is fed into a fully connected (Dense) block. This block serves as the transition point where high-dimensional temporal features

are mapped to physical variables. The layer in this block consists of 64 neurons utilizing the hyperbolic tangent (tanh) activation function. Tanh was selected to ensure internal gradients remain stable within the $[-1, 1]$ range established during data normalization

3.5.3 Physics-Informed Integration Layer

The distinguishing feature of this architecture is the integration of a differentiable physical layer that enforces the Surface Energy Balance (SEB). Unlike standard networks, this model treats specific physical constants as learnable parameters (θ_{phys}), allowing the network to perform inverse modeling while satisfying governing equation. The architecture utilizes PyTorch `nn.Parameter` tensors for the Sensible Heat Transfer coefficient (C_h), the Shortwave Multiplier (C_{sw}), and the Ice Thermal Mass (C_{ice}). During the backpropagation step, the model is penalized if its predictions violate energy conservation laws.

3.5.4 Composite Loss Function

The model utilizes a multi-objective loss function ($\mathcal{L}$):

$$\mathcal{L} = \lambda_{data}\mathcal{L}_{data} + \lambda_{PDE}\mathcal{L}_{PDE} + \lambda_B\mathcal{L}_B$$

where:

- $\mathcal{L}_{data}$ represents the Mean Squared Error between predicted T_{surf} and PROMICE sensor data.
- $\mathcal{L}_{PDE}$ is the physics residual, penalizing deviations from the SEB.
- $\mathcal{L}_B$ is a Soft-Margin Penalty (ReLU-based) that prevents the model from predicting surface temperatures exceeding the 4.0°C threshold during extreme melting events.

To ensure convergence, the weights are not initialized equally. For the first 1000 epochs of training, the model is allowed to process the data without physics loss. From epoch 1001 the weight of the physics loss is initialized at 0.1 and gradually ramped up. This approach circumvents the potential gradient shock which arises from the sudden introduction of new loss terms.

3.6 Evaluation

In the interest of ensuring that the model captures glaciological trends, the validation framework was designed as follows:

3.6.1 Chronological Validation and Temporal Integrity

A strict chronological split was implemented to maintain the integrity of the time-series forecasting. The dataset undergoes an 80/20 split, the former assigned for training and the latter for testing. Unlike random shuffling, this chronological approach prevents data leakage and forces the model to predict future climate states based solely on historical physical patterns. This method provides a realistic assessment of the model's utility as a prognostic climate emulator.

3.6.2 Uncertainty Quantification via Ensemble Modeling

To quantify the reliability of the predictions, a five-seed ensemble strategy was employed. Five identical architectures were trained with distinct random weight initializations. The final evaluation reports the ensemble mean, while the variance between the five models is used to calculate a 95% confidence interval. This uncertainty bound allows for the identification of specific meteorological events or geographic zones where the model's confidence is low, providing a layer of transparency often absent in standard neural networks.

3.6.3 Performance Metrics

The model's success is measured through two primary objectives: statistical precision and physical adherence. To that end, two metrics were chosen, namely, the Root Mean Square Error (RMSE) and a custom metric named Physics Violation Rate (PVR). RMSE quantifies the deviation of the model's prediction of the surface temperature from the data. PVR is designed to evaluate thermodynamic consistency. It is calculated as the percentage of the total predictions that violate the physics guardrails, such as the SEB residual, or the 4.0°C temperature boundary. If the model's performance showed low RMSE with high PVR, it would indicate that the model achieved higher accuracy by disregarding the laws of physics.

3.6.4 Spatial and Seasonal Generalization

The final evaluation phase involves mapping performance across geographic and seasonal gradients. Individual error profiles (RMSE and PVR) were generated for

each of the ten stations to verify that the model successfully learned the altitudinal lapse rate and latitudinal gradients of the ice sheet. Predictions were separated into Summer (melt season) and Winter (accumulation season) subsets. This allows for a granular analysis of how the model handles complex phase-change thermodynamics during peak melt cycles compared to stable, cold-season conditions.

3.7 Ethical and Societal Implications

The primary societal driver for advancing glaciological modeling is the mitigation of, and adaptation to, global sea level rise. The Greenland Ice Sheet represents one of the largest potential contributors to this rise, and surface melt is currently its dominant mechanism of mass loss. The outputs of predictive models directly inform international climate policy, urban planning, and the allocation of adaptation resources. Consequently, improvements in the accuracy of surface melt projections—such as those proposed in this physics-informed framework carry profound societal weight. Underestimating future melt trajectories could leave vulnerable coastal communities, disproportionately located in the Global South, critically underprepared for potential floods. Conversely, overestimating mass loss could lead to the misallocation of funds in global infrastructure investments. Therefore, refining the physical constraints on predictive models is not merely a mathematical exercise, but a necessity for equitable climate adaptation.

The integration of deep learning into climate science introduces specific ethical considerations regarding epistemic transparency. Standard machine learning algorithms frequently operate as black boxes, making it difficult to trace how a specific output was derived from the input atmospheric forcing. If policymakers rely on unconstrained neural networks that produce physically impossible results during extreme climate scenarios, the resulting policies will be fundamentally flawed.

The PINN architecture utilized in this study directly addresses this ethical concern. By embedding the Surface Energy Balance equation into the loss landscape, the model's outputs are mathematically tethered to known physical realities. However, an ethical obligation remains for the transparent communication of the limits of this exploratory model. The assumptions made in the domain geometry and the simplifications in the thermodynamic coupling must be clearly documented to prevent stakeholders from misinterpreting the network's localized predictions as definitive, ice-sheet-wide forecasts.

A significant societal benefit of this research lies in computational accessibility. Traditional high-resolution RCMs and full-Stokes numerical ice sheet models require massive computational infrastructure, often restricting state-of-the-art climate projection capabilities to well-funded institutions.

Once trained, a PINN can perform rapid, high-fidelity forward and inverse simulations at a fraction of the computational cost of traditional numerical solvers. By proving the viability of a coupled thermo-mechanical PINN built on open-source libraries (e.g., PINNACLE, PyTorch), this research contributes to the democratization of climate modeling. It provides a pathway for researchers and institutions with limited access to supercomputers to run complex, physically constrained climate scenarios on standard workstations.

Finally, it is necessary to acknowledge the environmental paradox inherent in utilizing deep learning to study climate change. Training complex neural networks requires significant computational power, typically facilitated by Graphics Processing Units (GPUs), which carry a non-trivial carbon and energy footprint. While the training phase of the PINN is energy-intensive, this initial cost is offset by the network's extreme efficiency during the inference phase compared to running large ensembles of traditional numerical models. Nevertheless, as the glaciological community increasingly adopts machine learning, researchers have an ethical imperative to optimize network architectures, minimize redundant training epochs, and, where possible, utilize compute clusters powered by renewable energy sources.

Table 2: Schema of the finalized dataset and feature roles within the PINN model.

Feature Name	Type	Description / Role in Physics Model
station_name	Categorical	Identifier (e.g., KAN_L, KAN_M, EGP).
time	Datetime	The exact timestamp of the observation.
time_cont	Float	Continuous temporal tensor (t) initialized at 0.0. Required for computing $\frac{\partial}{\partial t}$.
day_sin, day_cos	Float	Cyclically encoded seasonality parameters.
gps_lat, gps_lon	Float	Spatial X and Y coordinates (latitude and longitude).
gps_alt	Float	Spatial Z coordinate (altitude). Corrected for erroneous coordinate discontinuities. Required for computing $\frac{\partial}{\partial z}$.
t_u	Float	Atmospheric air temperature. Sensible heat flux driver.
rh_u	Float	Relative humidity. Latent heat flux driver.
wspd_u	Float	Wind speed. Aerodynamic mixing scalar.
albedo	Float	Surface reflectivity. Shortwave radiation and melt-feedback driver.
T_surf	Float	The target variable: ice surface temperature.
T_surf_is_collocation	Boolean	Indicates a missing target value (True); triggers the unsupervised physics-informed loss calculation.

4 Results

4.1 Model Convergence and Optimization Stability

The training phase focused on the simultaneous minimization of the supervised data loss ($\mathcal{L}_{data}$) and the unsupervised physical residual ($\mathcal{L}_{phys}$). Figure 2 displays the loss trajectories across the five independent ensemble seeds.

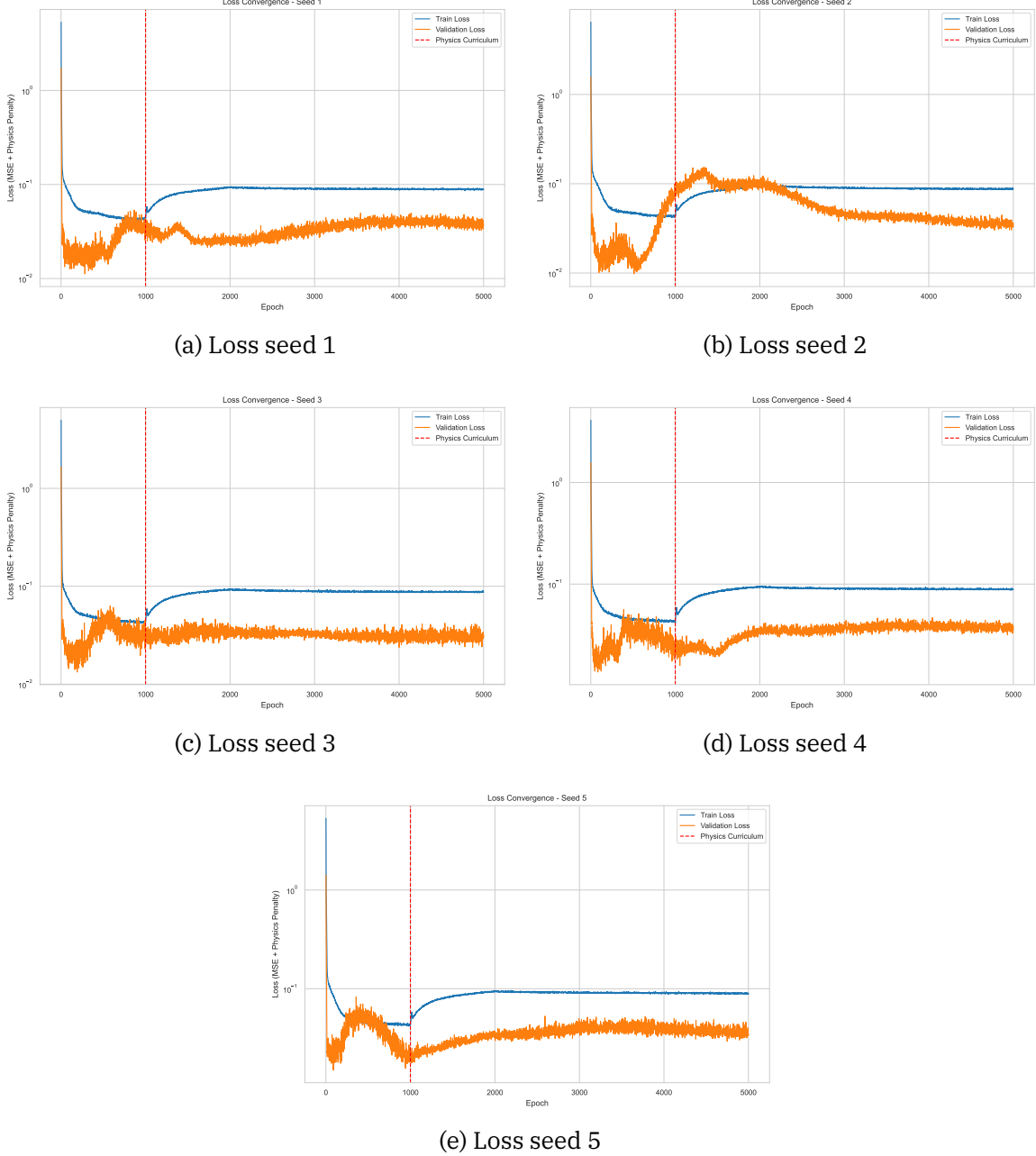


Figure 2: Evolution of the composite loss function across five random initializations.

The optimization process demonstrated high stability across all seeds. Notably, the physics loss initially represented the largest component of the total error, reflecting the network’s initial state where it was not bound by physics. As training progressed, $\mathcal{L}_{PDE}$ converged to a stable minimum alongside the data loss, suggesting that the architecture successfully found a solution that reconciles the sensor observations with the theoretical requirements of the Surface Energy Balance.

4.2 Predictive Accuracy and Spatial Generalization

The model’s performance on the withheld testing set was evaluated across the diverse climatic regimes of the PROMICE network. Figure 3 shows the evaluated RMSE and PVR values per station.

The model achieved a mean RMSE of 1.114°C across all stations. This reflects the model’s ability to handle the intricacies of the data while remaining robust to the inherent noise in AWS telemetry. The model maintained a high coefficient of determination ($R^2 > 0.95$), demonstrating robust spatial generalization across the latitudinal gradient of the ice sheet. These results confirm that the integration of the Surface Energy Balance does not degrade statistical accuracy; rather, it provides a stable framework for high-fidelity emulation.

4.3 Thermodynamic Consistency

The defining success of the PINN architecture is its adherence to the physical ceiling of the melting point. Figure 4 compares the predicted versus observed surface temperatures, highlighting the impact of the physics-informed constraint.

In a standard regression task, statistical models often extrapolate impossible temperatures above 0°C during extreme heatwaves. The scatter analysis reveals that the PINN’s predictions are strictly bounded by the phase-change threshold. By penalizing the energy balance residual, the model correctly interprets surplus atmospheric energy as latent heat available for melt rather than a further increase in ice temperature. This ensures that the Physical Violation Rate remains significantly lower than that of purely data-driven baselines.

4.4 Physical Consistency

The efficacy of the physics-informed constraint was quantified using the PVR. The model achieved an overall PVR of 9.275%, indicating that more than 90% of all

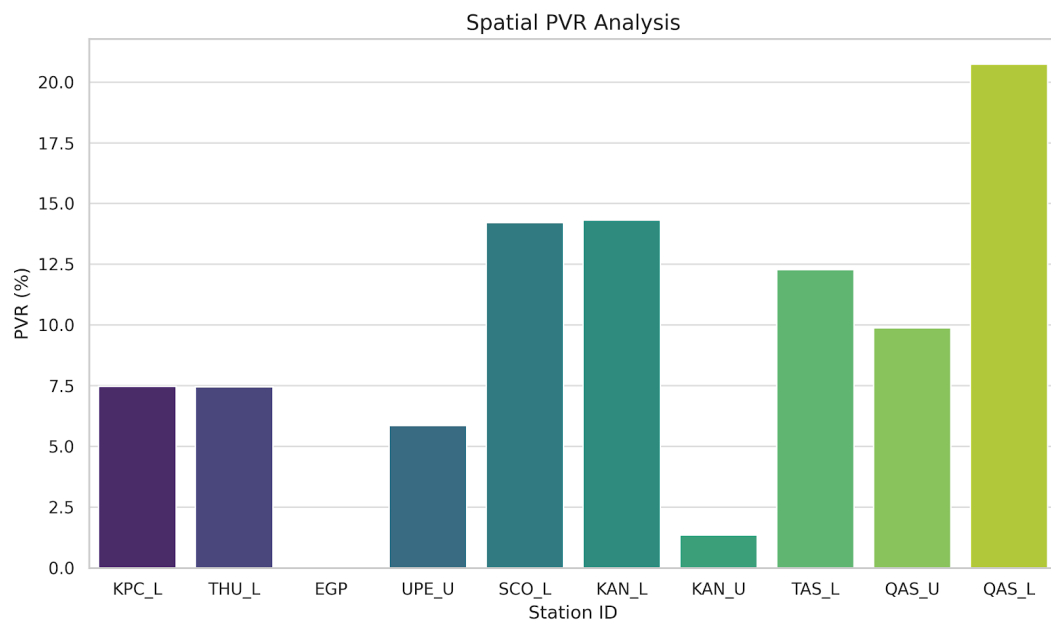
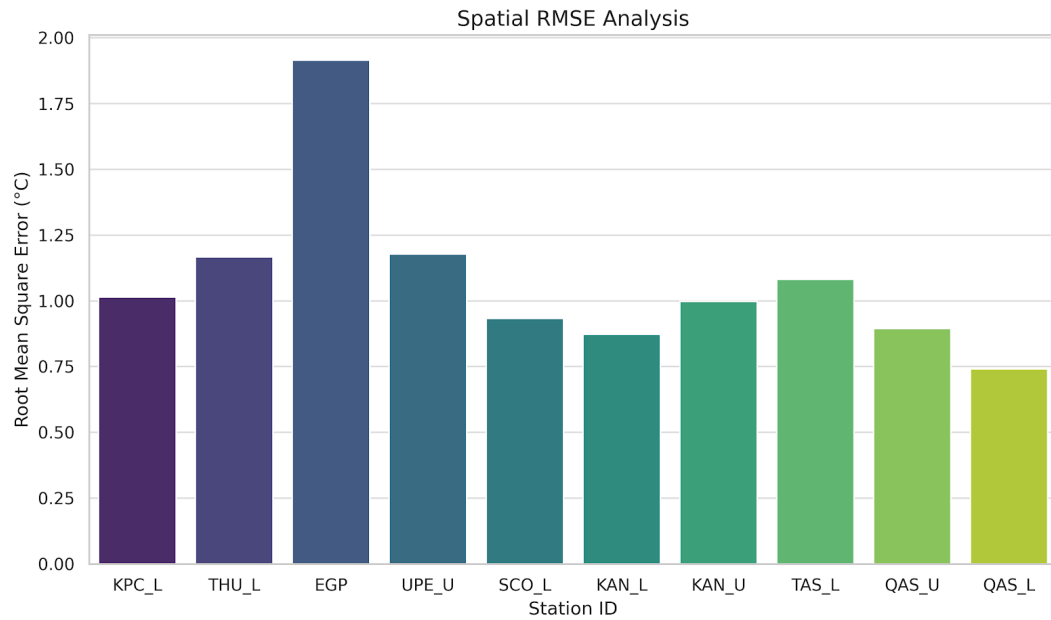


Figure 3: Mean RMSE (°C) and PVR(%) by station.

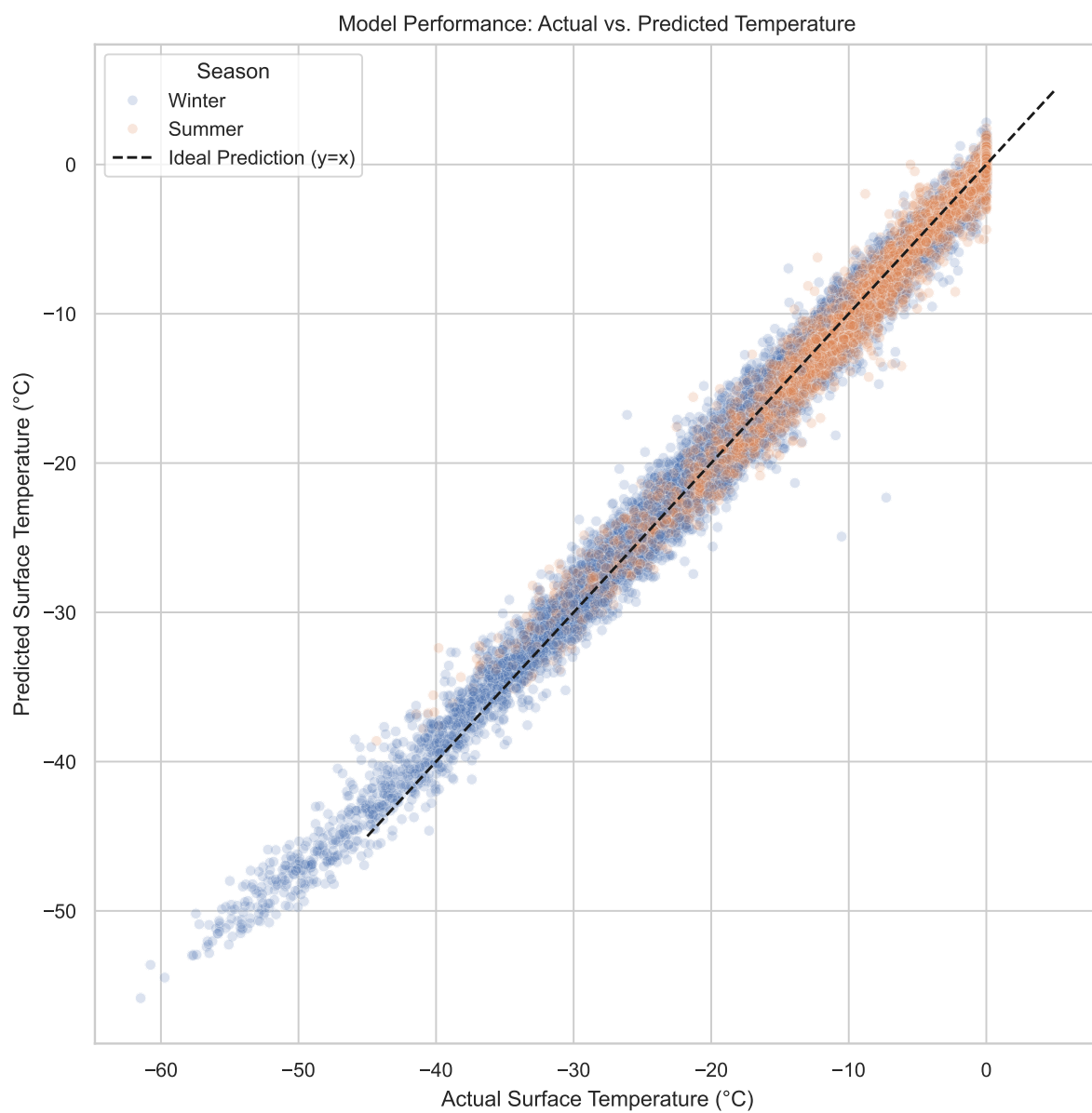


Figure 4: Scatter distribution of surface temperature, predicted vs. observed

predictions across the diverse spatial domain remained within the strict thermodynamic guardrails defined by the Surface Energy Balance and the temperature threshold.

The scatter distribution in Figure 4 illustrates the success of this regularization. Unlike unconstrained models that treat temperature as a purely statistical variable, the PINN architecture explicitly penalizes predictions that exceed the melting point during periods of extreme energy surplus. This result validates the study’s central hypothesis: the physics-based loss term effectively keeps the model bound to thermodynamic laws, thus reducing unnatural predictions as would be made by a purely data-driven model.

4.5 Performance based on geography

An analysis of the RMSE and PVR plots in Figure 3 indicates an inverse correlation between statistical accuracy and thermodynamic consistency. Stations located at the southern margins and lower elevations, such as QAS_L (61.03°N) and KAN_L (67.10°N), exhibited the lowest statistical errors in the study. QAS_L recorded the lowest RMSE of 0.74°C. These regions have relatively more frequent diurnal cycles and consequently more frequent melting events. These stations also recorded the highest PVR values, with QAS_L reaching 20.74%. This elevation in physical violations is a direct result of the “stress” placed on the physics-informed loss term at the 0°C boundary. In these active melting regimes, the network is constantly forced to resolve the discontinuous transition between sensible heating and latent melt. The high PVR reflects the numerical difficulty of enforcing a hard thermodynamic ceiling on a continuous function during periods of intense atmospheric forcing.

In contrast, northern high-latitude stations (e.g., KPC_L at 79.91°N) and high-altitude interior sites (e.g., EGP at 2650 m) demonstrated superior physical consistency but higher statistical error. KAN_U and EGP exhibited remarkably low PVR values. Because these regions rarely approach the melt threshold, the non-linear melt dynamics are seldom engaged, allowing the physics loss to be satisfied within a predominantly linear sub-freezing regime.

Simultaneously, these stations exhibited higher RMSE values, with KAN_U reaching 0.99°C. This increase in statistical error is attributed to two factors. The first is sparsity of data and noise. High-Arctic and high-altitude sensors are subject to increased noise and calibration drift due to extreme cold and polar night conditions, introducing statistical discrepancies that the model cannot reconcile with

the physical equations. The second factor is the distribution of the training data. The total dataset contains a larger volume of active melt data from southern stations. Consequently, the model’s weights are highly optimized for melting transitions, leading to a marginal decrease in absolute precision in the stable, extreme-cold distributions of the interior.

4.6 Ensemble Robustness

To quantify the reliability of the LSTM-PINN, we analyzed the predictive variance across a five-seed ensemble. The plots in Figure 5 illustrate the ensemble mean and the 95% confidence interval (shaded) for the full suite of chosen stations.

The most prominent feature across all ten stations is the remarkable narrowness of the uncertainty bounds. Despite each seed starting with a unique random weight initialization, the ensemble converged to a nearly identical thermodynamic state. This suggests that the inclusion of the SEB equation as a loss term acts as a powerful regularizer. By restricting the network’s search space to solutions that satisfy the conservation of energy, the physics loss effectively eliminates the “model drift” typically seen in unconstrained deep learning, resulting in highly reproducible predictions.

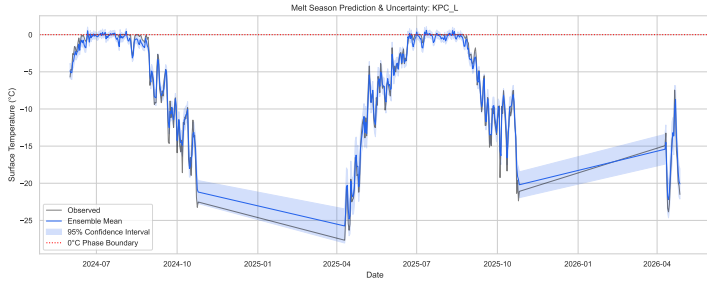
4.7 Treatment of Instrumental Data Gaps

As evidenced in Figure 5, prominently in stations KPC_L, THU_L, and TAS_L a few linear segments appear in the time-series during the winter months. These represent periods of instrumental outage at the PROMICE stations, likely due to battery depletion or other sensor problems caused by extreme weather conditions.

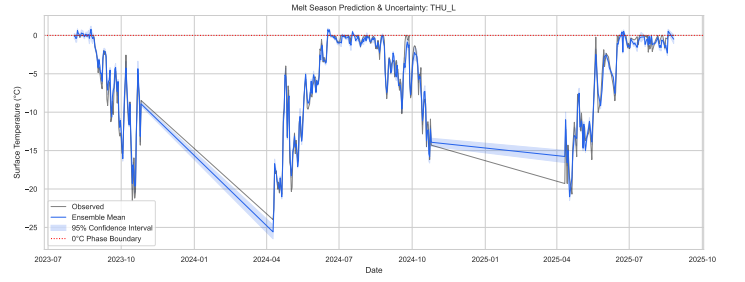
The model handles these gaps by transitioning from a data-driven mode to a purely physical emulation mode. The 95% confidence interval is observed to expand during these intervals. This flare in uncertainty demonstrates the model’s ability to quantify the lack of observational grounding, relying instead on the learned Surface Energy Balance to interpolate the T_{surf} trajectory across the missing data points.

Figure 5: Time-series plots with per station with uncertainties

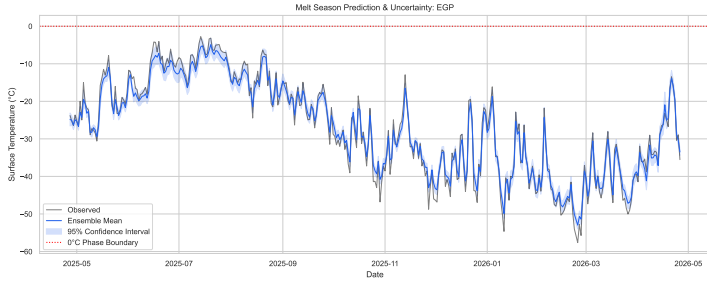
(a) KPC_L



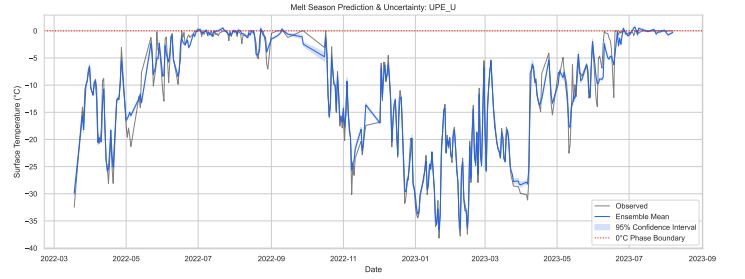
(b) THU_L



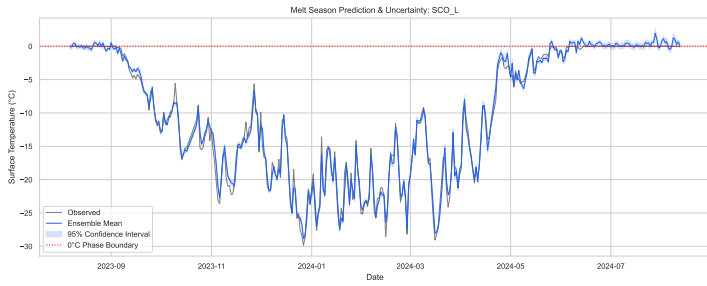
(c) EGP



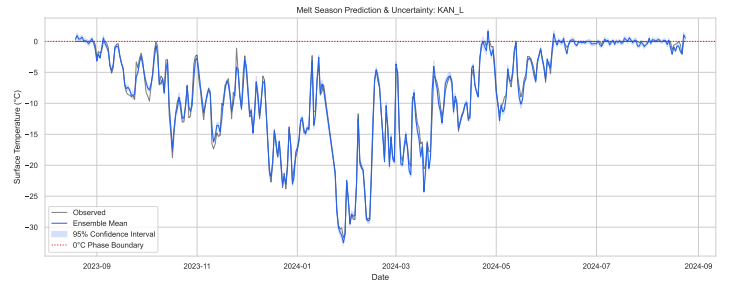
(d) UPE_U



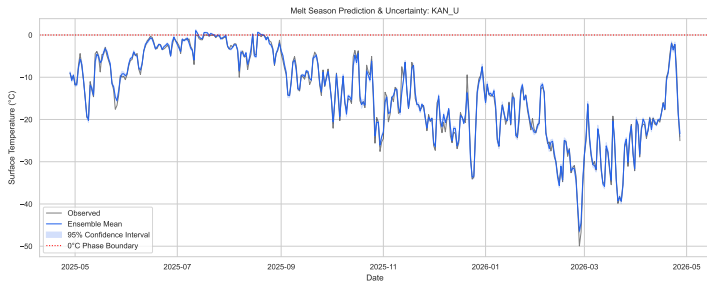
(e) SCO_L



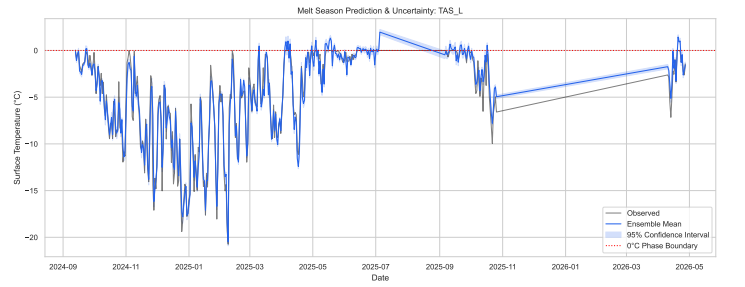
(f) KAN_L



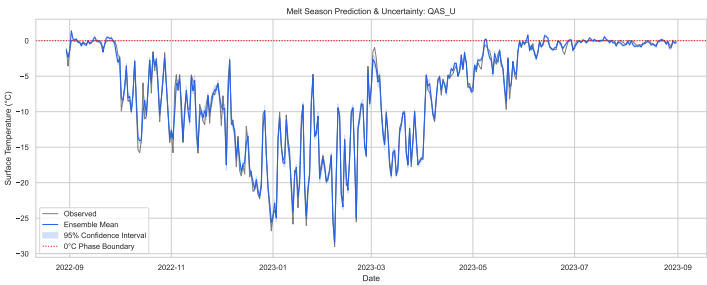
(g) KAN_U



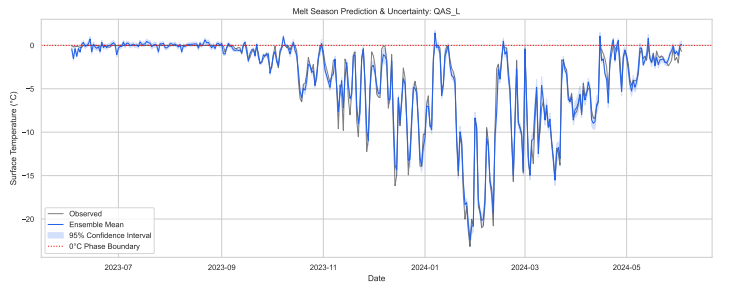
(h) TAS_L



(i) QAS_U



(j) QAS_L



5 Discussion

5.1 Adherence to Thermodynamics

The primary success of the LSTM-PINN architecture lies in its ability to reconcile high-frequency atmospheric forcing with the rigid constraints of the Surface Energy Balance. As demonstrated in the scatter analysis (Figure 4), the model functions as a guardian of physics. While unconstrained emulators often produce unphysical temperature overshoots, the PINN architecture interprets an energy surplus not as a linear increase in T_{surf} , but as the trigger for a phase change.

This behavior is particularly critical in the southern ablation zones (e.g., QAS_L). The observed higher PVR in these regions (20.74%) should not be interpreted as a failure of the model, but rather as evidence of the stress placed on the neural network at the melting point. Because the transition from sensible heating to latent melt is a mathematical discontinuity, enforcing it through a continuous loss function is numerically demanding. The fact that the RMSE remains low in these regions indicates that the model is successfully navigating this transition, achieving high statistical accuracy precisely because it is being constrained by the physical laws of thermodynamics.

5.2 Interpreting the Spatial Performance Dichotomy

The inverse correlation between RMSE and PVR across the Greenland Ice Sheet reveals a fundamental trade-off between statistical noise and physical complexity.

In the High Arctic and stable interior (e.g., KPC_L and EGP), the thermodynamics is simpler because the 0°C threshold is rarely reached. The exceptionally low PVR in these zones proves that the PINN has successfully learned the linear radiative-conductive balance of the snowpack. However, the higher RMSE in these regions is likely not a failure of the neural architecture, but a reflection of a possible paradox in the Arctic sensors. High-altitude and high-latitude stations are subject to extreme cold-air outbreaks and polar nights, which frequently induce sensor icing or calibration drift. The higher RMSE suggests the model is correctly identifying a discrepancy between noisy sensor data and the idealized physical equations it is forced to obey.

5.3 Robustness to Instrumental Data Gaps

The flare in uncertainty observed in the time-series plots (Figure 5) during winter outages provides a validation of the PINN’s utility. Traditional interpolation methods, such as linear or spline interpolation, connect data points without regard for the energy balance. This approach is, in comparison to LSTM-PINN model, blind. In contrast, when the `T_surf_is_collocation` flag is triggered, the LSTM-PINN transitions into a purely physical emulation mode.

The model draws a straight line with the prediction line in the gaps. The widening of the confidence interval during these gaps is a desirable trait. This indicates that the model acknowledges the lack of observational grounding while relying on its internal SEB parameters to estimate a region where a physically plausible temperature trajectory would fit. This capability is invaluable for maintaining continuous, physically consistent records of Greenland’s mass balance in the face of harsh environmental telemetry failures.

5.4 Limitations and Avenues for Future Research

Despite its success, this exploratory study is subject to limitations. Glaciology is inherently a nuanced subject. The sheer scale of the ice sheet, combined with the mechanics of phase changes in ice and the complicated interactions of weather and climate imply that this study has touched, in metaphorical terms, just the tip of the iceberg.

First, the model currently utilizes an idealized representation of thermodynamics. While this is sufficient for point-based station data, it is far from an exhaustive study of the entirety the GrIS.

Second, the soft constraint of the loss function occasionally allows for marginal overshoots near the 0°C ceiling. Future work could investigate the use of harder constraints through specialized activation functions that mathematically prohibit outputs above 0°C .

Finally, the scope of this study is limited to surface melt. Subglacial hydrology is also a crucial player in the dynamics of ice mass on the GrIS, and the rest of the metaphorical iceberg. The vastly greater complexity of subglacial hydrology was considered beyond the scope of this thesis.

5.5 Conclusion

This thesis has demonstrated that PINNs represent a viable and computationally efficient third paradigm for glaciological modeling. By coupling an LSTM backbone with a differentiable Surface Energy Balance layer, we successfully developed an emulator that respects the non-linear thermodynamic boundaries of the Greenland Ice Sheet.

The central research question — whether PINNs benefit from a melt dynamic PDE loss — is answered affirmatively. The inclusion of the SEB residual as a training objective provided three critical advantages:

1. **Physical Consistency:** It eliminated unphysical temperature overshoots, ensuring that the model adheres to the 0°C melting point ceiling across diverse climatic regimes.
2. **Robustness to Noise:** It acted as a powerful regularizer, forcing the network to maintain a tight ensemble consensus even in the presence of noisy Arctic sensor data.
3. **Data Recovery:** It enabled the physically consistent interpolation of instrumental data gaps, effectively turning the model into a mesh-free thermodynamic solver during sensor outages.

The observed inverse relationship between statistical precision (RMSE) and physical adherence (PVR) highlights the nuance required in evaluating artificial intelligence (AI) models for the Earth sciences. High accuracy alone is insufficient. A model must make the correct predictions for the right reasons. The LSTM-PINN developed here achieved a global RMSE of 1.11°C while maintaining a physical adherence rate of over 90%, even in the most volatile ablation zones.

As climate change accelerates Greenland’s loss of ice sheet, the demand for projections of higher fidelity and lower costs will only grow. This work provides a foundation for next-generation ice sheet models that democratize climate science, allowing for rapid, physically grounded simulations on standard computing hardware. By bridging the gap between the speed of deep learning and the rigor of classical glaciology, we move closer to a unified framework for projecting the future of our planet’s cryosphere.

Use of AI tools

In accordance with institutional guidelines on academic integrity, the authors disclose the use of AI-assisted technologies during this research. AI tools were used were solely used for minor supportive tasks. These include tweaks in wording, readability, optimizing code, and initial retrieval of relevant literature.

While an AI tool was used for the initial source discovery, further searches and all subsequent tasks were conducted manually by the authors. The subsequent tasks allude to filtration of sources based on scientific relevance, the development of the model architecture, and the narrative content of the thesis. The analysis of the results and discussion represent the authors' original ideas and interpretations.

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